Experiment No. 3

Aim : Write Shell Scripts Using Control Structures such as Conditional and Looping Statement.

PROBLEM STATEMENTExecute andimplement shell scripting using conditional statements (if, if-else) and looping statements (for, while) in Linux.

Theory

A Shell Script is a text file containing a sequence of Linux commands. It helps automate repetitive tasks and improves system administration efficiency.

Conditional Statements

Conditional statements execute commands based on a condition.

```
f [ condition ]
then
   commands
else
   commands
fi
```

Looping Statements

A loop executes a set of commands repeatedly.

For Loop

Used when the number of iterations is known.

```
for variable in list
do
  commands
```

done

While Loop

Used when the number of iterations is not fixed.

```
while[ condition ]
do
  commands
done
```

Linux Commands Used

Command	Purpose
nano	Create/Edit file
Chmod + x	Make script executable
./filename	Execute script
echo	Display output
read	Accept input

Applications

* User input validation
* Menu-driven programs
* Automation of repetitive tasks
* System administration
* File management

Precautions

* Start the script with!#bin/bash.

* Save the file withsh extension.

* Giveexecutepermission before running.

* Check syntax carefully.

Result

Thus, the shell script using conditional and looping statements was written and executed successfully.

Viva Questions

1. What is Shell Scripting?

2. What is Bash?

3. What is the use of #/bin/bash?

4. What isan ifstatement?

5. What isa forloop?

6. What is a while loop?

7. What is the purpose of read?

8. Whatis chmod +x?

9. How do you execute a shell script?

10. What is the difference between `if` and `if-else`?